

ADVANCING RELIABILITY AND MEDICAL DATA ANALYSIS THROUGH NOVEL STATISTICAL DISTRIBUTION EXPLORATION

BRODERICK OLUYEDE* — LEON SCHRÖDER** — SEAN FANG*** — ACHRAF COHEN**
— THATAYAONE MOAKOFI* — YUHAO ZHANG**** — SHUSEN PU**, c

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ABSTRACT. This comprehensive study delves into the examination and application of novel statistical distributions, namely the Ristić-Balakrishnan-Topp-Leone-Exponentiated half Logistic-G (RB-TL-EHL-G) family of distributions, emphasizing their paramount importance in reliability and medical data modeling. We meticulously explore a multitude of this family of novel distributions, accentuating their respective features, properties, and real-world applicability. The probability density, the cumulative distribution, the hazard rate, and the quantile functions are provided. The density functions of the RB-TL-EHL-G family are expanded, enabling a deeper understanding of their statistical properties, including various moments, generating functions, order statistics, stochastic orderings, probability weighted moments, and the Rényi entropy. A significant portion of the investigation is dedicated to the intensive analysis of various data sets, to which these distributions are fitted, unveiling noteworthy insights into their behavior and performance. Furthermore, the discussions extend to a comparative study, delineating the advantages and limitations of each distribution, fostering a deeper understanding and selection criteria for practitioners.

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1. Introduction

The evolution of data landscapes prompts a burgeoning necessity for the development of novel probability distributions. This innovation broadens the palette of distributions at the disposal of researchers, thereby enhancing our ability to encapsulate the complexities inherent in real-world phenomena. It also provides avenues to surmount limitations ingrained in existing models, enriching our understanding and the application of statistical analysis across a variety of domains.

The array of statistical distributions delineated in existing literature has found resonance in myriad sectors including, but not limited to, engineering, finance, actuarial sciences, demography, reliability, and economics. However, the scope of behaviors these distributions can portray is somewhat circumscribed, often falling short of providing comprehensive solutions to all practical scenarios. This underscores the exigent call for the engendering of new distributions and models.

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B. Oluyede, L. Schröder and S. Pu contributed equally to this work.

c Corresponding author.

A slew of distributions, conceived as enhancements to existing models, have been introduced in recent years, including exponentiated flexible Lomax distribution [13], type II exponentiated half-logistic Gompertz-G family of distributions [16], the Ristić-Balakrishnan Topp-Leone-Gompertz-G family of distributions [21], the gamma odd power generalized Weibull-G family of distributions [9], the Alpha power Rayleigh-G family of distributions [1], the odd exponential-logarithmic family of distributions [6], a modified power family of distributions [8], the gamma-Weibull-G Family of distributions [20], and the odd Chen-G family of distributions [3].

In this paper, we will introduce another family of distributions that excel at modeling a range of hazard rate functions, including those with rising and falling shapes. These instances also exhibit adeptness in depicting light, elongated, and truncated data sets, delivering superior fits compared to standard and numerous expanded distributions documented hitherto. A facet of intrigue is the role played by the additional shape parameter(s), which instills skewness and modulates the heaviness of the tails in any established distribution.

This manuscript unfolds as follows: Section 2 delineates the cdf, pdf, hazard rate, and quantile functions. Section 3 elucidates the RB-TL-EHL-G pdf linear representation. In Section 4, derivations of the statistical and mathematical properties are presented. Three special cases are spotlighted in Section 5. Section 6 unveils the maximum likelihood estimation method, while Monte Carlo simulation results showcasing precision are unveiled in Section 7. Section 8 unfolds the fitting distribution applications utilizing real-life data, culminating with the conclusion in Section 9.

2. Ristić-Balakrishnan-Topp-Leone-Exponentiated Half Logistic-G Family of Distributions

Based on the Ristić-Balakrishnan-G [24] and Topp-Leone-Exponentiated half logistic-G families of distributions [19], we propose a novel family of distributions, named the Ristić-Balakrishnan-Topp-Leone-exponentiated half logistic-G (RB-TL-EHL-G) family of distributions. Its cumulative distribution function (cdf) is given by

$$F_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi) = 1 - \frac{1}{\Gamma(\sigma)} \int_0^{-\log \left(\left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)} t^{\sigma-1} e^{-t} dt$$

$$= 1 - \frac{\gamma \left(\sigma, -\log \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)}{\Gamma(\sigma)}.$$
(2.1)

Corresponding to equation (2.1) the probability density function (pdf) is

$$f_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi) = \frac{4ab}{\Gamma(\sigma)} \left(-\log \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)^{\sigma-1}$$

$$\times \frac{g(x; \Psi) (G(x; \Psi))^{a-1}}{[1 + \overline{G}(x; \Psi)]^{a+1}}$$

$$\times \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2 \right]^{b-1} \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right),$$
(2.2)

for $\sigma, b, a > 0$ and parameter vector Ψ .

2.1. Hazard Rate and Quantile Functions

Let the random variable X be continuous with pdf $f(x)$, cdf $F(x)$ and survival function $\bar{F}(x) = 1 - F(x)$. Then the hazard rate function (hrf), mean residual life function, and reverse hazard rate function (rhf) of X are defined as $h_F(x) = \frac{f(x)}{F(x)}$, $\delta_F(x) = \int_x^\infty \frac{\bar{F}(u)}{\bar{F}(x)} du$ and $\tau_F(x) = \frac{f(x)}{F(x)}$. [26] showed that $h_F(x)$, $\tau_F(x)$ and $\delta_F(x)$ behave equivalently. In this paper, we solely present the hazard rate function. Given the pdf (equation (2.2)) and cdf (equation (2.1)), the hazard rate function of the RB-TL-EHL-G Family of distributions is

$$\begin{aligned} h_{RB-TL-EHL-G}(x) &= \frac{4abg(x; \Psi) (G(x; \Psi))^{a-1}}{[1 + \bar{G}(x; \Psi)]^{a+1}} \left(-\log \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \bar{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)^{\sigma-1} \\ &\times \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \bar{G}(x; \Psi)} \right)^a \right)^2 \right]^{b-1} \left(1 - \left(\frac{G(x; \Psi)}{1 + \bar{G}(x; \Psi)} \right)^a \right) \\ &\times \frac{1}{\gamma(\sigma, -\log \left(\left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \bar{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right))}. \end{aligned}$$

In order to obtain the quantile function, we need to solve for the inverse cdf as

$$F_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi) = 1 - \frac{\gamma \left(\sigma, -\log \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \bar{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)}{\Gamma(\sigma)} = p$$

for $0 \leq p \leq 1$. Therefore, it suffices to solve

$$G(x; \Psi) = \frac{2 \left(1 - \sqrt{1 - \exp \left(-\frac{\gamma^{-1}[\Gamma(\sigma)(1-p), \sigma]}{b} \right)} \right)^{\frac{1}{a}}}{1 - \left(1 - \sqrt{1 - \exp \left(-\frac{\gamma^{-1}[\Gamma(\sigma)(1-p), \sigma]}{b} \right)} \right)^{\frac{1}{a}}} := r. \quad (2.3)$$

Consequently, the calculation for the quantile function becomes simpler:

$$x_r = G^{-1}(r), \quad (2.4)$$

where r is given in equation (2.3).

3. Expansion of the Density Functions

In this section, the series expansions for the RB-TL-EHL-G density functions are presented. Consider the following series representation $(-\log(1-q)) = \sum_{i=0}^{\infty} \frac{q^{i+1}}{i+1}$, and let $q = \left(1 - \left(\frac{G(x; \Psi)}{1 + \bar{G}(x; \Psi)} \right)^a \right)^2$, then

$$\left(-\log \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \bar{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)^{\sigma-1} = b^{\sigma-1} \left[\sum_{m,s=0}^{\infty} \binom{\sigma-1}{m} b_{s,m} q^{m+s+\sigma-1} \right]. \quad (3.1)$$

Equations(3.1) were derived in [10, 22], where $b_{s,m} = (sc_0)^{-1} \sum_{l=1}^s [m(l+1) - s] c_l b_{s-l,m}$, $b_{0,m} = c_0^m$, and $c_s = (s+2)^{-1}$. The pdf of RB-TL-EHL-G family of distribution (equation (2.2)) can therefore be expressed as

$$\begin{aligned} f_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi) &= \frac{4ab^\sigma}{\Gamma(\sigma)} \left[\sum_{m,s,k=0}^{\infty} \binom{\sigma-1}{m} \binom{b-1}{k} (-1)^k b_{s,m} q^{m+s+\sigma+k-\frac{1}{2}} \right] \\ &\times g(x; \Psi) (G(x; \Psi))^{a-1} [1 + \bar{G}(x; \Psi)]^{-a-1}. \end{aligned} \quad (3.2)$$

Substituting q back into equation (3.2), we obtain

$$\begin{aligned} & f_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi) \\ &= \frac{4ab^\sigma}{\Gamma(\sigma)} \left[\sum_{m,s,k,j,i,l,r=0}^{\infty} \binom{\sigma-1}{m} \binom{b-1}{k} \binom{2m+2s+2\sigma+2i-1}{j} \binom{a+aj-1}{i} \right. \\ & \quad \times \left. \binom{-a-aj-1}{l} \binom{i+l}{r} (-1)^{k+j+i+r} b_{s,m} g(x; \Psi) [G(x; \Psi)]^r \right]. \end{aligned} \quad (3.3)$$

To simplify the notation, we let

$$\begin{aligned} w_r &= \sum_{m,s,k,j,i,l=0}^{\infty} \frac{4ab^\sigma}{\Gamma(\sigma)} \binom{\sigma-1}{m} \binom{b-1}{k} \binom{2m+2s+2\sigma+2i-1}{j} \binom{a+aj-1}{i} \\ & \quad \times \binom{-a-aj-1}{l} \binom{i+l}{r} \frac{(-1)^{k+j+i+r}}{r+1} b_{s,m}. \end{aligned} \quad (3.4)$$

Finally, we can express the pdf of RB-TL-EHL-G family of distributions as

$$f_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi) = \sum_{r=0}^{\infty} w_r (r+1) g(x; \Psi) [G(x; \Psi)]^r = \sum_{r=0}^{\infty} w_r g_{E-G}(x; \Psi, r+1), \quad (3.5)$$

where the pdf of the exponentiated-G (Exp-G) distribution with power parameter $r+1$ is denoted by $g_{E-G}(x; \Psi, r+1)$.

Thus, using the Exp-G distributions [12, 18], we may derive the key statistical characteristics of the RB-TL-EHL-G family of distributions.

4. Mathematical and Statistical Properties

In this section, you will find comprehensive details regarding the RB-TL-EHL-G family of distributions. This includes their moment generating function, incomplete and conditional moments, order statistics, probability weighted moments, and the Rényi entropy. Along this section, X and Y_{r+1} refer to random variables with a density function $f_{RB-TL-EHL-G}(x)$ as in equation (2.2) and an exponentiated-G distribution with power parameter $(r+1)$.

4.1. Moments and Generating Functions

The RB-TL-EHL-G family of distributions' n^{th} moment is calculated as follows:

$$E(X^n) = \int_{-\infty}^{\infty} x^n f_{RB-TL-EHL-G}(x) dx = \sum_{r=0}^{\infty} w_r E(Y_{r+1}^n), \quad (4.1)$$

where w_r is provided in equation (3.4), and $E(Y_{r+1}^n)$ defines n^{th} moment of the exponentiated-G distribution with power parameter $(r+1)$. Moreover, for $s < 1$, the moment generating function (mgf) is

$$M_X(s) = \sum_{r=0}^{\infty} w_r M_{r+1}(s), \quad (4.2)$$

where w_r is described in equation (3.4) and $M_{r+1}(s)$ is the mgf of Y_{r+1} .

4.2. Incomplete and Conditional Moments

Inequality measures such as Lorenz and Bonferroni curves and lifetime models frequently use conditional and incomplete moments. One can find the n^{th} incomplete moment of X as

$$\xi_n(t) = \int_{-\infty}^t x^n f_{RB-TL-EHL-G}(x) dx = \sum_{r=0}^{\infty} w_r \int_{-\infty}^t x^n g_{r+1}(x; \Psi) dx, \quad (4.3)$$

where $g_{r+1}(x; \Psi)$ represents the pdf of the Exp-G distribution with power parameter $(r+1)$. We get the RB-TL-EHL-G family of distributions' first incomplete moment by fixing $n = 1$ in equation (4.3). The RB-TL-EHL-G family of distributions' n^{th} conditional moment is defined by

$$E(X^n | X \geq a) = \frac{1}{\bar{F}_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi)} \sum_{r=0}^{\infty} w_r E(Y_{r+1}^n I_{Y_{r+1}^n \geq t}), \quad (4.4)$$

with $\bar{F}_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi) = 1 - F_{RB-TL-EHL-G}(x; \sigma, b, a, \Psi)$ and

$$E(Y_{r+1}^n I_{Y_{r+1}^n \geq t}) = (r+1) \int_{G(z; \Psi)}^1 [Q_G(z; \Psi)]^n z^{r+1} dz, \quad (4.5)$$

with the parameter vector Ψ . Let F have the vitality function $V_F(a)$. Then $E(X - a | X > a) = E(X | X > a) - a + V_F(a) - a$ is the formula for the mean residual life function. Using the incomplete and conditional moments, it is easy to get the mean deviations, Lorenz and Bonferroni curves.

4.3. Order Statistics

Several fields, including auction modeling, industrial process optimization, distribution parameter estimation and actuarial science, use order statistics. Assume that $X_1, \dots, X_n$ are i.i.d. RB-TL-EHL-G random variables, and $F(x)$ and $f(x)$ the cdf and pdf of the RB-TL-EHL-G family of distributions. The pdf of the k^{th} order statistic $f_{k:n}(x)$ can be written as follows:

$$f_{k:n}(x) = \frac{n!}{(k-1)!(n-1)!} \times \sum_{r=0}^{\infty} w_r (r+1) g(x; \Psi) [G(x; \Psi)]^r \left[\sum_{r=0}^{\infty} w_r [G(x; \Psi)]^{r+1} \right]^{i+j-1}. \quad (4.6)$$

Here we may write (using the result of Gradshteyn and Ryzhik [10]),

$$\left[\sum_{r=0}^{\infty} w_r [G(x; \Psi)]^{r+1} \right]^{i+j-1} = \sum_{r=0}^{\infty} d_{u,i+j-1} [G(x; \Psi)]^{r+1}, \quad (4.7)$$

where $d_{0,i+j-1} = (w_0)^{i+j-1}$,

$$d_{u,i+j-1} = [(u+1)w_0]^{-1} \sum_{s=0}^u [s(i+j) - u - 1] w_s d_{u-s,i+j-1}, \quad (4.8)$$

for $u = 1, 2, \dots$ and w_s is given in equation (3.4). Then

$$f_{k:n}(x) = \sum_{j=0}^{n-i} \sum_{r,u=0}^{\infty} s_{j,u,r} g_{r+u+1}(x; \Psi), \quad (4.9)$$

where

$$s_{j,u,r} = \frac{n! w_u d_{u,i+j-1} (u+1)}{(i-1)!(n-i)!(r+u+1)} (-1)^j \binom{n-i}{j}, \quad (4.10)$$

and $g_{r+u+1}(x; \Psi)$ is the exponentiated-G distribution with parameter $(r+u+1)$. As a result, we are able to obtain the characteristics of the RB-TL-EHL-G distribution from those of the distribution of the order statistic from the Exp-G family.

4.4. Stochastic Orderings

Stochastic order, hazard rate order and likelihood ratio order for the RB-TL-EHL-G family of distributions are all presented in this subsection. The necessary definitions can be found in [21] and [27].

Consider X_1 and X_2 as two independent random variables following RB-TL-EHL-G family of distributions with $X_1 \sim F_1(x; \sigma_1, b, a, \Psi)$ and $X_2 \sim F_2(x; \sigma_2, b, a, \Psi)$, then

$$\frac{f_1(x)}{f_2(x)} = \frac{\Gamma(\sigma_2)}{\Gamma(\sigma_1)} \left(-\log \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)^{\sigma_1 - \sigma_2}. \quad (4.11)$$

Differentiating equation (4.11) with respect to x yields

$$\begin{aligned} \frac{d}{dx} \left(\frac{f_1(x)}{f_2(x)} \right) &= \frac{-ab^{\sigma_1 - \sigma_2}(\sigma_1 - \sigma_2)\Gamma(\sigma_2)}{\Gamma(\sigma_1)} \left(-\log \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)^{\sigma_1 - \sigma_2 - 1} \\ &\quad \times \frac{2 \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)}{1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2} \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^{a-1} \frac{2g(x; \Psi)}{(1 + \overline{G}(x; \Psi))^2} \end{aligned} \quad (4.12)$$

which is negative if $\sigma_1 > \sigma_2$. We conclude that X_1 and X_2 are stochastically ordered because the likelihood ratio order $X \preceq_{lr} Y$ exists.

4.5. Probability Weighted Moments

The probability weighted moments (PWMs) [11], which are used to describe distributions, are

$$M_{i,j,k} = E \left(X^i F^j(X) [1 - F(X)]^k \right) = \int_{-\infty}^{\infty} x^i F^j(x) [1 - F(x)]^k f(x) dx,$$

where i, j and k are real numbers and the distribution of X is given by $F \equiv F(x) = P(X \leq x)$. The PWM of order (i, j, k) is proportional to the i^{th} moment around the origin of the $(j+1)^{th}$ order statistic for the sample of size $n = j + k + 1$, when j and k are non-negative integers. Let X be a random variable with a distribution in the RB-TL-EHL-G family, then

$$F^j(x) [1 - F(x)]^k f(x) = \sum_{h=0}^k \binom{k}{h} (-1)^h \left[\sum_{r=0}^{\infty} w_r [G(x; \Psi)]^{r+1} \right]^{j+h} \left[\sum_{r=0}^{\infty} w_r (r+1) g(x; \Psi) [G(x; \Psi)]^r \right],$$

and apply a similar formula in equation (4.7), we have

$$F^j(x) [1 - F(x)]^k = \sum_{h=0}^k \sum_{r,u=0}^{\infty} s_{h,r,u} g_{r+u+1}(x; \Psi), \quad (4.13)$$

where

$$s_{h,r,u} = \frac{w_r(r+1)d_{r,j+h}}{r+u+1} \binom{k}{h} (-1)^h, \quad d_{0,j+h} = (w_0)^{j+h}, \quad (4.14)$$

$$d_{r,j+h} = [(r+1)w_0]^{-1} \sum_{t=1}^r [t(j+h+1) - r - 1] w_t d_{r-t,j+h}, \quad (4.15)$$

for $r = 1, 2, \dots$, and w_r given in equation (3.4). Hence, the PWMs of the RB-TL-EHL-G family of distributions can be expressed as

$$M_{i,j,k} = \sum_{h=0}^k \sum_{r,u=0}^{\infty} s_{h,r,u} \int_{-\infty}^{\infty} x^i g_{r+u+1}(x; \Psi) dx, \quad (4.16)$$

that is, the moments of the exponentiated G-distribution can be used to calculate the PWMs of the RB-TL-EHL-G family of distributions.

4.6. Rényi Entropy

Rényi entropy [23] is used in information theory that generalizes a number of associated entropies including min-, Hartley, collision and Shannon entropy. In domains like ecology and statistics, Rényi entropy has a significant role as a gauge of diversity. By letting $f_{RB-TL-EHL-G}(x) = f(x)$. The RB-TL-EHL-G family of distributions' Rényi entropy is provided by

$$I_R(p) = \frac{1}{1-p} \log \left(\int_0^\infty f^p(x) dx \right), \quad p > 0 \text{ and } p \neq 1, \quad (4.17)$$

where

$$\begin{aligned} f^p(x) &= \left(\frac{4ab}{\Gamma(\sigma)} \right)^p \frac{g^p(x; \Psi) (G(x; \Psi))^{p(a-1)}}{[1 + \overline{G}(x; \Psi)]^{p(a+1)}} \left(-\log \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2 \right]^b \right)^{p(\sigma-1)} \\ &\times \left[1 - \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2 \right]^{p(b-1)} \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^p. \end{aligned} \quad (4.18)$$

Let $y = \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2$. Following a similar procedure as in the expansion of the pdf (equation (3.1)), we can show that equation (4.18) can be expressed as

$$f^p(x) = \left(\frac{4^p a^p b^{p\sigma}}{\Gamma^p(\sigma)} \right) \left[\sum_{m,s,n=0}^{\infty} \binom{p\sigma-p}{m} \binom{pb-p}{n} (-1)^n b_{s,m} y^{m+s+p\sigma-\frac{p}{2}+n} \right] \frac{g^p(x; \Psi) (G(x; \Psi))^{p(a-1)}}{[1 + \overline{G}(x; \Psi)]^{p(a+1)}}.$$

Note that with $y = \left(1 - \left(\frac{G(x; \Psi)}{1 + \overline{G}(x; \Psi)} \right)^a \right)^2$, then the RB-TL-EHL-G family of distributions' Rényi entropy can be expressed as

$$\begin{aligned} I_R(p) &= \frac{1}{1-p} \log \left\{ \frac{4^p a^p b^{p\sigma}}{\Gamma^p(\sigma)} \sum_{m,s,n,j,k=0}^{\infty} \binom{p\sigma-p}{m} \binom{pb-p}{n} \binom{2(m+s+p\sigma+n)-p}{j} \right. \\ &\times \sum_{l=0}^{\infty} \binom{-p(a+1)-aj}{k} \binom{k}{l} (-1)^{n+j+l} b_{s,m} \frac{1}{\left(1 + \frac{p(a-1)+aj+l}{p} \right)^p} \\ &\times \int_0^\infty \left[\left(1 + \frac{p(a-1)+aj+l}{p} \right) g(x; \Psi) (G(x; \Psi))^{\frac{p(a-1)+aj+l}{p}} \right]^p dx \Big\} \\ &= (1-p)^{-1} \log \left(\sum_{l=0}^{\infty} C_l e^{(1-p)I_{REG}} \right), \end{aligned} \quad (4.19)$$

where $I_{REG} = (1-p)^{-1} \int_0^\infty \left[\left(1 + \frac{p(a-1)+aj+l}{p} \right) g(x; \Psi) (G(x; \Psi))^{\frac{p(a-1)+aj+l}{p}} \right]^p dx$ is the Rényi entropy of the exponentiated-G distribution with power parameter $\left(1 + \frac{p(a-1)+aj+l}{p} \right)$ and the weights are

$$\begin{aligned} C_l &= \frac{4^p a^p b^{p\sigma}}{\Gamma^p(\sigma)} \sum_{m,s,n,j,k=0}^{\infty} \binom{p\sigma-p}{m} \binom{pb-p}{n} \binom{2(m+s+p\sigma+n)-p}{j} \\ &\times \binom{p(a+1)+aj}{k} \binom{k}{l} (-1)^{n+j+l} b_{s,m} \frac{1}{\left(1 + \frac{p(a-1)+aj+l}{p} \right)^p}. \end{aligned} \quad (4.20)$$

The Rényi entropy of the RB-TL-EHL-G family of distributions may then be calculated from that of the exponentiated-G distribution using equation (4.19).

5. Some Special Cases of the RB-TL-EHL-G Family of Distributions

When the baseline cdf $G(x; \Psi)$ is specified, various probability distributions are generated. In this section, we will present a few special models.

5.1. RB-TL-EHL-Log-Logistic (RB-TL-EHL-LLoG) Distribution

Given the cdf and pdf of the log-logistic distribution as $G(x; c) = 1 - (1 + x^c)^{-1}$ and $g(x; c) = cx^{c-1}(1 + x^c)^{-2}$ for $c > 0$ and $x > 0$, we define the cdf and pdf of RB-TL-EHL-Log-Logistic distribution as follows

$$F(x; \sigma, b, a, c) = 1 - \frac{\gamma \left(\sigma, -\log \left[1 - \left(1 - \left(\frac{1 - (1 + x^c)^{-1}}{1 + (1 + x^c)^{-1}} \right)^a \right)^2 \right]^b \right)}{\Gamma(\sigma)}$$

and

$$f(x; \sigma, b, a, c) = \frac{4ab}{\Gamma(\sigma)} \left(1 - \left(\frac{1 - (1 + x^c)^{-1}}{1 + (1 + x^c)^{-1}} \right)^a \right) \left(-\log \left[1 - \left(1 - \left(\frac{1 - (1 + x^c)^{-1}}{1 + (1 + x^c)^{-1}} \right)^a \right)^2 \right]^b \right)^{\sigma-1} \\ \times \left[1 - \left(1 - \left(\frac{1 - (1 + x^c)^{-1}}{1 + (1 + x^c)^{-1}} \right)^a \right)^2 \right]^{b-1} \frac{cx^{c-1}(1 + x^c)^{-2} (1 - (1 + x^c)^{-1})^{a-1}}{[1 + (1 + x^c)^{-1}]^{a+1}},$$

respectively, for $\sigma, b, a, c > 0$.

Figure 1 displays a few examples of the RB-TL-EHL-Log-Logistic distribution. The pdf may take numerous shapes, such as nearly symmetric, J, left-skewed, and right-skewed. In addition, hrf graphs for the RB-TL-EHL-Log-Logistic distribution display monotonically increasing, monotonically decreasing, upside-down bathtub followed by bathtub, and inverted bathtub shapes.

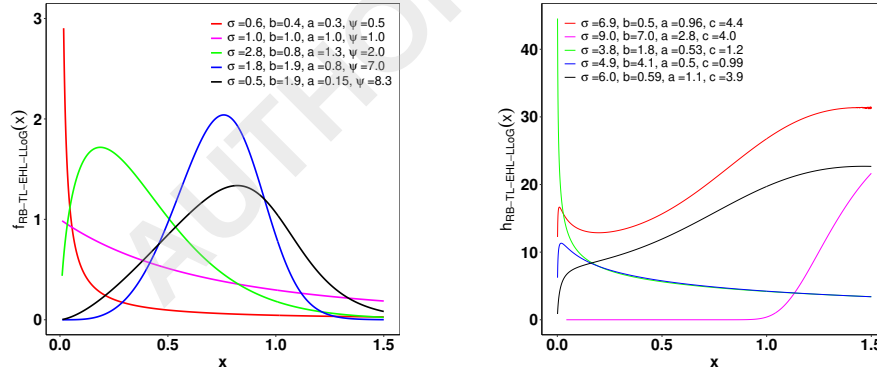


FIGURE 1. A) The pdf of RB-TL-EHL-Log-Logistic for different values of parameters σ , b , a , and c . B) The hrf of RB-TL-EHL-Log-Logistic for different values of parameters σ , b , a , and c .

5.2. RB-TL-EHL-Weibull (RB-TL-EHL-W) Distribution

Suppose the cdf and pdf of the Weibull distribution are given by $G(x; \gamma) = 1 - \exp(-x^\gamma)$ and $g(x; \gamma) = \gamma x^{\gamma-1} \exp(-x^\gamma)$, for $\gamma > 0$, and $x > 0$. Therefore, the cdf and pdf of the RB-TL-EHL-W distribution are given by

$$F(x; \sigma, b, a, \gamma) = 1 - \frac{\gamma \left(\sigma, -\log \left[1 - \left(1 - \left(\frac{1 - \exp(-x^\gamma)}{1 + \exp(-x^\gamma)} \right)^a \right)^2 \right]^b \right)}{\Gamma(\sigma)}$$

and

$$f(x; \sigma, b, a, \gamma) = \frac{4ab}{\Gamma(\sigma)} \left(-\log \left[1 - \left(1 - \left(\frac{1 - \exp(-x^\gamma)}{1 + \exp(-x^\gamma)} \right)^a \right)^2 \right] \right)^{\sigma-1} \left(1 - \left(\frac{1 - \exp(-x^\gamma)}{1 + \exp(-x^\gamma)} \right)^a \right) \\ \times \left[1 - \left(1 - \left(\frac{1 - \exp(-x^\gamma)}{1 + \exp(-x^\gamma)} \right)^a \right)^2 \right]^{b-1} \frac{\gamma x^{\gamma-1} \exp(-x^\gamma) (1 - \exp(-x^\gamma))^{a-1}}{[1 + \exp(-x^\gamma)]^{a+1}},$$

respectively, for $\sigma, b, a, \gamma > 0$.

Figure 2 portrays the probability density function (pdf) and hazard rate function (hrf) plots for the RB-TL-EHL-W distribution under varying parameter values. The pdf can adopt various configurations, including right-skewed, left-skewed, unimodal, and J-shaped. Contrastingly, the RB-TL-EHL-W hazard demonstrates an extensive variety of forms such as increasing, decreasing, inverted bathtub, inverted bathtub succeeded by a regular bathtub, and a sequence of a regular bathtub, followed by an inverted one, and concluding with another regular bathtub.

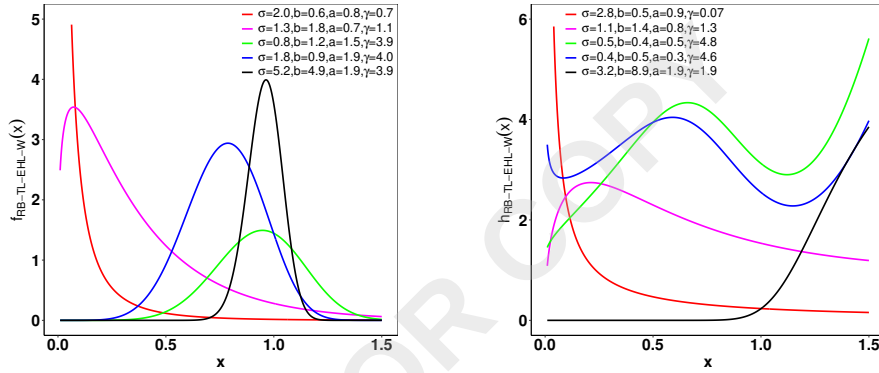


FIGURE 2. A) The pdf of RB-TL-EHL-W for different values of parameters σ , b , a , and γ . B) The hrf of RB-TL-EHL-Weibull for different values of parameters σ , b , a , and γ .

6. Maximum Likelihood Estimation

We will be using the maximum likelihood estimation (MLE) technique to estimate the parameters of the RB-TL-EHL-G family of distributions in this section. Assume that the independent random sample $(X_1, X_2, \dots, X_n)$ is obtained from the RB-TL-EHL-G family of distributions with its model parameters vector $\Delta = (\sigma, b, a, \Psi)^T$. The log-likelihood function $l_n = \log(L_n(\Delta))$ from the obtained values takes the form:

$$l_n = n \ln \left(\frac{4ab}{\Gamma(\sigma)} \right) + (\sigma - 1) \sum_{i=1}^n \ln \left(-\log \left[1 - \left(1 - \left(\frac{G(x_i; \Psi)}{1 + \overline{G}(x_i; \Psi)} \right)^a \right)^2 \right] \right)^b \\ + (b - 1) \sum_{i=1}^n \ln \left[1 - \left(1 - \left(\frac{G(x_i; \Psi)}{1 + \overline{G}(x_i; \Psi)} \right)^a \right)^2 \right] + \sum_{i=1}^n \ln \left(1 - \left(\frac{G(x_i; \Psi)}{1 + \overline{G}(x_i; \Psi)} \right)^a \right) \\ + \sum_{i=1}^n \ln [g(x_i; \Psi)] + (a - 1) \sum_{i=1}^n \ln [G(x_i; \Psi)] - (a + 1) \sum_{i=1}^n \ln [1 + \overline{G}(x_i; \Psi)]. \quad (6.1)$$

We numerically maximize the log-likelihood function $\log(L_n(\Delta))$ in order to get the maximum likelihood estimates. This may be achieved by concurrently solving the nonlinear system of equations $(\frac{\partial l_n}{\partial \sigma}, \frac{\partial l_n}{\partial b}, \frac{\partial l_n}{\partial a}, \frac{\partial l_n}{\partial \Psi})^T = 0$. Due to the fact that these equations are not in closed form, the MLEs can

be obtained by numerically maximizing $\log(L_n(\Delta))$ with respect to the parameters using a numerical technique like the Newton-Raphson method. The Supplementary Information provides the partial derivatives of the log-likelihood function with respect to each element of the parameter vector.

7. Simulation Results

We conducted a Monte Carlo simulation study to evaluate the accuracy of maximum likelihood estimates for the RB-TL-EHL-LLoG distribution, which is a part of the RB-TL-EHL-G family of distributions. Our simulation involved generating 3000 random samples from the RB-TL-EHL-LLoG distribution, with sample sizes of 25, 50, 100, 200, 400, and 800.

To evaluate the accuracy of the estimates, the mean estimate (Mean), average bias (ABias), and root mean square error (RMSE) were computed. Specifically, for the estimated parameter $\hat{\lambda}$, the ABias and RMSE were calculated as follows, respectively:

$$ABias(\hat{\lambda}) = \frac{1}{N} \sum_{i=1}^N (\hat{\lambda}_i - \lambda), \quad \text{and} \quad RMSE(\hat{\lambda}) = \sqrt{\frac{\sum_{i=1}^N (\hat{\lambda}_i - \lambda)^2}{N}},$$

Based on the data presented in Tables 1 and 2, it can be inferred that the RB-TL-EHL-LLoG distribution's maximum likelihood estimates (MLEs) are reliable. This is because the mean values closely match the actual parameter values, and both RMSE and ABias decrease with larger sample sizes for all parameters. To evaluate the performance of various estimation methods, including MLE, Least Squares (LS) and Weighted Least Squares (WLS), Maximum Product Spacing (MPS), Cramér-von Mises (CVM), and Anderson-Darling Estimation (ADE), we employed a standardized procedure to generate random samples for each method. The formulations for these estimations are thoroughly detailed in our previous works [5, 25, 29]. The simulation results for the selected parameter sets are presented in Figure 3. As depicted in the figure, MLE and ADE consistently converge more rapidly compared to the other methods, whereas LS and CVM generally exhibit slower convergence.

TABLE 1. Monte Carlo Simulation Results

(0.4, 0.4, 0.9, 2.5)						(0.4, 0.9, 1.2, 1.0)		
Parameter	Sample Size	Mean	RMSE	ABias		Mean	RMSE	ABias
σ	25	1.9423	8.5374	1.5423		0.8877	2.6104	0.4877
	50	0.5953	1.1799	0.1953		0.5536	0.5288	0.1536
	100	0.4499	0.1971	0.0499		0.4750	0.2051	0.0750
	200	0.4199	0.1082	0.0199		0.4443	0.1448	0.0443
	400	0.4107	0.0741	0.0107		0.4507	0.1287	0.0507
	800	0.4040	0.0523	0.0040		0.3898	0.0847	-0.0101
b	25	2.0724	6.1320	1.6724		1.8302	1.9788	0.9302
	50	1.5949	4.1457	1.1949		1.7631	1.8191	0.8631
	100	1.0220	1.8296	0.6220		1.4380	1.4076	0.5380
	200	0.7767	1.3466	0.3767		1.1930	0.9403	0.2930
	400	0.5603	0.5058	0.1603		1.0742	0.6786	0.1742
	800	0.4873	0.3205	0.0873		0.9874	0.4394	0.0874
a	25	1.4406	1.4856	0.5406		1.5890	1.5422	0.3890
	50	1.2230	0.9972	0.3230		1.4235	1.0776	0.2235
	100	1.0495	0.6930	0.1495		1.3028	0.7373	0.1028
	200	0.9915	0.4747	0.0915		1.2510	0.4887	0.0510
	400	0.9498	0.3360	0.0498		1.2365	0.3514	0.0365
	800	0.9246	0.2385	0.0246		1.2275	0.2625	0.0275
c	25	1.9403	1.6013	-0.5596		2.0161	2.1235	1.0161
	50	2.1343	1.3290	-0.3656		1.8007	1.7680	0.8007
	100	2.3908	1.0359	-0.1091		1.4985	1.3902	0.4985
	200	2.4861	0.7528	-0.0138		1.3100	1.0033	0.3100
	400	2.5087	0.5692	0.0087		1.1592	0.6489	0.1592
	800	2.5071	0.2784	0.0071		1.0743	0.4060	0.0743

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TABLE 2. Monte Carlo simulation results

Parameter	Sample Size	(0.6, 0.6, 1.2, 2.1)			(1.1, 1.5, 0.4, 2.5)		
		Mean	RMSE	ABias	Mean	RMSE	ABias
σ	25	1.5569	1.9026	0.9569	1.9179	1.8133	0.8179
	50	1.2372	1.4577	0.6372	1.6180	1.4778	0.5180
	100	1.0060	1.1098	0.4060	1.3682	1.1713	0.2682
	200	0.7727	0.5942	0.1727	1.2751	0.9393	0.1751
	400	0.6728	0.3385	0.0728	1.1852	0.7086	0.0852
	800	0.6258	0.2112	0.0258	1.1596	0.5683	0.0596
b	25	1.3397	1.5875	0.7397	1.3191	1.0417	-0.1808
	50	1.1318	1.2449	0.5318	1.3562	0.8737	-0.1437
	100	0.9283	0.9187	0.3283	1.4346	0.5080	-0.0653
	200	0.8869	0.7491	0.2869	1.4854	0.3443	-0.0145
	400	0.7639	0.6004	0.1639	1.4905	0.2353	-0.0094
	800	0.6220	0.4546	0.0220	1.5016	0.1724	0.0016
a	25	2.0839	3.6768	0.8839	1.1508	1.7078	0.7508
	50	1.5811	2.5249	0.3811	0.8845	1.1501	0.4845
	100	1.2494	1.5754	0.0494	0.6616	0.6693	0.2616
	200	1.2450	0.8402	0.0450	0.5101	0.3396	0.1101
	400	1.2276	0.6089	0.0276	0.4491	0.2040	0.0491
	800	1.2244	0.4128	0.0244	0.4219	0.1385	0.0219
c	25	2.3907	1.3569	0.2907	2.3575	1.2104	-0.1424
	50	2.3085	1.0521	0.2085	2.3717	0.8983	-0.1282
	100	2.2739	0.8408	0.1739	2.4076	0.7168	-0.0923
	200	2.1989	0.5596	0.0989	2.4724	0.5369	-0.0275
	400	2.1482	0.3795	0.0482	2.4852	0.3786	-0.0147
	800	2.1285	0.2510	0.0285	2.5047	0.2831	0.0047

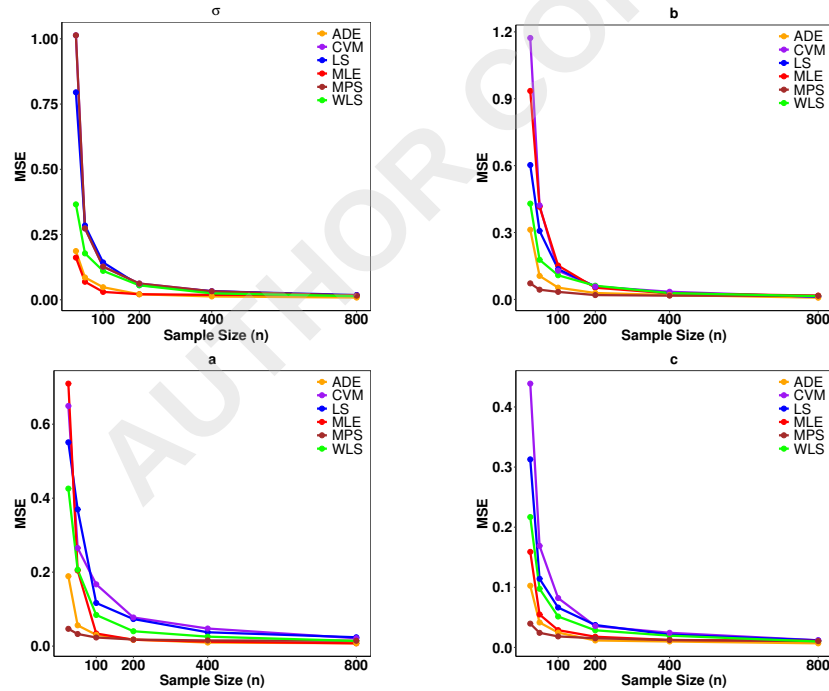


FIGURE 3. Mean squared error of parameter estimates for various methods, with parameters set to $\sigma = 0.4, b = 0.9, a = 0.4, c = 2.5$.

8. Real-World Applications

In this section, we focus on evaluating the suitability of the RB-TL-EHL-G family of distributions by fitting the RB-TL-EHL-LLoG distribution, its special case, to three diverse datasets. To compare its fit, we examine the distributions of the gamma exponentiated Lindley log-logistic (GELLoG) distribution

[17], exponentiated half-logistic odd Weibull-Topp-Leone-log logistic (EHLOW-TL-LLoG) distribution [7], odd exponentiated half-logistic-Burr XII (OEHL-BXII) distribution [2], alpha power extended log-logistic (APExLLD) distribution [28], and alpha power Topp-Leone Weibull (APTLW) distribution [4]. You can find the pdf's of these distributions in the Supplementary Information.

In order to compare the distributions that have been fitted, we will utilize established goodness-of-fit statistics such as the -2log-likelihood statistic ($-2\ln(L)$), Akaike Information Criterion ($AIC = 2p - 2\ln(L)$), Consistent Akaike Information Criterion ($CAIC = AIC + 2\frac{p(p+1)}{n-p-1}$), Bayesian Information Criterion ($BIC = p\ln(n) - 2\ln(L)$), (where n represents the number of observations and p represents the number of estimated parameters), Cramér-von Mises statistic (W^*), Anderson-Darling statistic (A^*), Kolmogorov-Smirnov (K-S) statistic, and its corresponding p-value. It is widely understood that lower values for all of these goodness-of-fit statistics (except for the K-S statistic p-value) indicate a better fit for the model when applied to the given data set.

8.1. Eruptions Data

There is a blowhole located in Kiama, NSW, Australia. The times in seconds between eruptions back in 1998 were recorded. The data can be found in the Supplementary Information.

Parameter estimates and goodness-of-fit measures: ($2\log(L)$, AIC , $AICC$, BIC , W^* , A^* , K-S) are reported in Table 3. We see that the RB-TL-EHL-LLoG distribution results in the smallest values for the goodness-of-fit measures and also has the highest p-value for the K-S statistic. Consequently, out of the given distributions, the RB-TL-EHL-LLoG distribution fits the data the best.

TABLE 3. MLEs and Goodness-of-Fit Statistics for Eruptions Data

Model	Estimates				Statistics							
	σ	a	b	c	$-2\log L$	AIC	$AICC$	BIC	W^*	A^*	K-S	p-value
RB-TL-EHL-LLoG	94.8270 (3.8543×10^{-04})	1.6018 (3.8667×10^{-02})	119.3500 (3.7621×10^{-04})	0.1250 (1.0038×10^{-02})	587.8252	595.8252	596.5032	604.4608	0.1088	0.7998	0.0967	0.5865
GELLoG	λ	c	α	δ	599.9101	607.9101	608.5881	616.5456	0.2086	1.3568	0.1369	0.1809
	0.0946 (0.0190)	1.2788 (0.4142)	0.003 (0.0032)	12.6659 (0.0057)								
EHLOW-TL-LLoG	b	β	δ	c	588.6800	596.6800	597.3580	605.3155	0.1132	0.8296	0.0983	0.5659
	13.5542 (0.2628)	0.1003 (0.1221)	8.7495 (3.6356)	2.1340 (2.3159)								
OEHL-BXII	α	λ	a	b	621.7598	629.7598	630.4377	638.3953	0.2440	1.6560	0.1444	0.1383
	0.3118 (2.5591×10^{-02})	4.3343×10^{-05} (2.1975×10^{-05})	4.9485 (2.1289×10^{-03})	0.4720 (2.2345×10^{-02})								
APExLLD	α	a	b	c	591.4674	599.4674	600.1453	608.1029	0.1668	1.1671	0.0972	0.5800
	2.3071×10^{06} (1.0000×10^{-07})	1.4194 (0.1453)	3.3085 (2.5287)	0.8865 (0.8229)								
APTLW	θ	α	β	λ	598.4024	606.4024	607.0803	615.0379	0.1905	1.2501	0.1221	0.2957
	0.6297 (0.2647)	15.9900 (0.0037)	1.0971 (0.1979)	0.0115 (0.0115)								

In Figure 4, plots of the fitted density as well as the histogram and the comparison of expected and observed probability plots are provided. The fitted Kaplan-Meier curve, empirical cumulative distribution function (ECDF) plot, total time on test (TTT) plot and the fitted hrf can be found in Figure 5. We observe that the distribution is very close to the Kaplan-Meier survival curve and the empirical cdf is close to the cdf of the proposed distribution. The non-monotonic hrf is demonstrated by the TTT plot.

8.2. Remission Times Data

The data provided by [15] pertains to the number of months that 128 bladder cancer patients spent in remission. The data can be found in the Supplementary Information.

Table 4 summarizes the MLEs of several models' parameters and reports different goodness-of-fit statistics and the K-S p-values. We see that the RB-TL-EHL-LLoG distribution results in the smallest values for the goodness-of-fit statistics and also has the highest K-S p-value among all fitted distributions. As a result, out of the given distributions, the RB-TL-EHL-LLoG distribution fits the data the best.

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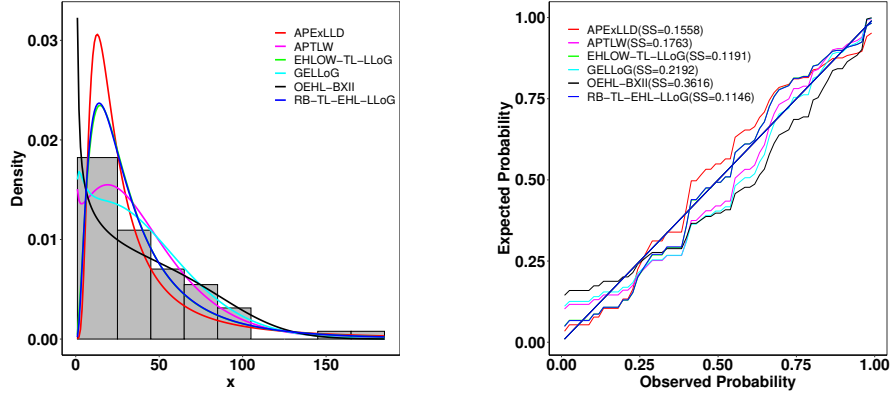


FIGURE 4. Histogram superposed by fitted density (left) and observed vs. expected probability plots (right) for the eruptions data

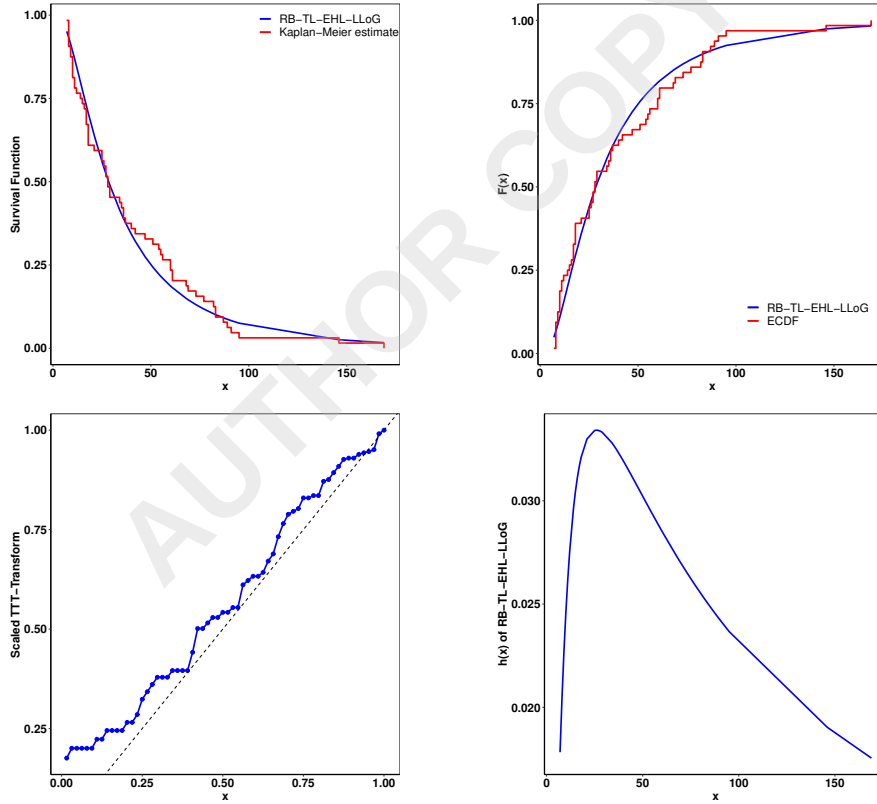


FIGURE 5. Fitted Kaplan-Meier (K-M) survival curve, theoretical and empirical cumulative distribution (ECDF), the total time on test (TTT) plot, and the hazard rate function (hrf) plot for the eruptions data

TABLE 4. MLEs and Goodness-of-Fit Statistics for Remission Times Data

Model	Estimates				Statistics							
	σ	a	b	c	$-2 \log L$	AIC	$AICC$	BIC	W^*	A^*	K-S	p-value
RB-TL-EHL-LLoG	23.5329 (0.5359)	3.9555 (1.4652)	8.7081 (3.6375)	0.2332 (0.0167)	827.9685	835.9685	836.2937	847.3766	0.0996	0.6807	0.0597	0.751
GELLoG	λ 0.1031 (0.0184)	c 0.6745 (0.0830)	α 15.4927 (18.8387)	δ 0.2613 (0.3252)	828.6728	836.6728	836.998	848.0809	0.1390	0.8477	0.0681	0.5917
EHLOW-TL-LLoG	b 14.8600 (6.0375)	β 0.7402 (0.1348)	δ 0.6542 (0.2358)	c 0.6177 (0.0682)	831.1801	839.1801	839.5053	850.5882	0.1455	0.9236	0.0882	0.2717
OEHL-BXII	α 0.3924 (0.0648)	λ 0.0274 (0.0187)	a 3.6828 (1.0211)	b 0.3659 (0.0975)	881.8121	889.8121	890.1373	901.2202	0.3274	1.9125	0.1165	0.0616
APExLLD	α 4.3332×10^{04} (1.3770×10^{-08})	a 0.9845 (0.0650)	b 4.1237×10^{-04} (2.4681×10^{-04})	c 7.8712×10^{02} (2.8809×10^{-08})	859.889	867.889	868.2142	879.2971	0.4889	3.0759	0.0968	0.1814
APTLW	θ 0.3632 (0.1239)	α 2.6272×10^{02} (4.8767×10^{-05})	β 0.8610 (0.1241)	λ 0.1120 (0.0540)	827.4507	835.4507	835.7759	846.8588	0.1275	0.7620	0.0727	0.5065

In Figure 6, plots of the fitted density as well as the histogram and the comparison of expected and observed probability plots are provided. The fitted Kaplan-Meier curve and CDF in Figure 7 are very close to the Kaplan-Meier survival curve and the empirical cdf. The non-monotonic hrf is demonstrated by the TTT plot.

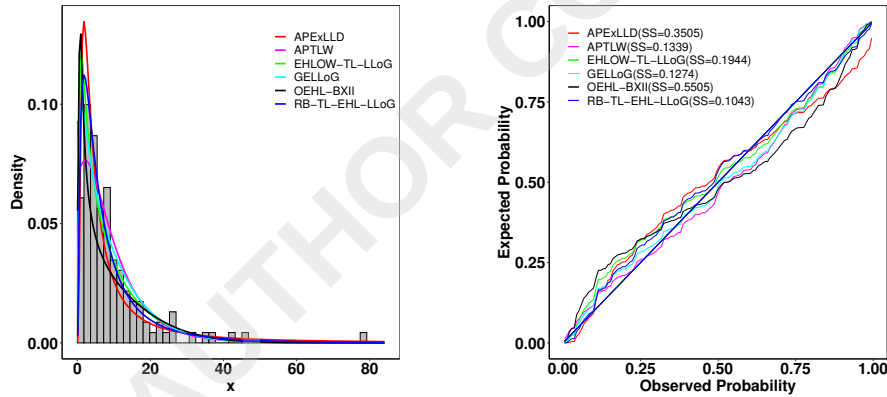


FIGURE 6. Histogram superposed by fitted density (left) and observed vs. expected probability plots (right) for the remission times data

8.3. Active Repair Times Data

The data set, which shows active repair times for airborne communication transceivers, was published by [14]. The data is in the Supplementary Information.

Parameter estimates and goodness-of-fit statistics are summarized in Table 5. We see that the RB-TL-EHL-LLoG distribution yields the smallest values for the goodness-of-fit statistics and also has the highest p-value for the K-S statistic. Consequently, out of the given distributions, the RB-TL-EHL-LLoG distribution fits the data the best.

Figure 8 provides plots of the fitted density as well as the histogram and the comparison of expected and observed probability plots. The fitted Kaplan-Meier curve, CDF, total time on test plot and the fitted hrf can be found in Figure 9. We observe that the distribution is very close to the Kaplan-Meier survival curve and the empirical cdf is close to the cdf of the proposed distribution. The non-monotonic hrf is demonstrated by the TTT plot.

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TABLE 5. MLEs and Goodness-of-Fit Statistics for Active Repair Times Data

Model	Estimates				Statistics							
	σ	a	b	c	$-2 \log L$	AIC	$AICC$	BIC	W^*	A^*	K-S	p-value
RB-TL-EHL-LLoG	73.2450 (2.3702×10^{-05})	0.5144 (6.5080×10^{-03})	389.4500 (5.3288×10^{-06})	0.1129 (0.0123)	181.457	189.457	190.5998	196.2125	0.0680	0.4822	0.1192	0.6200
GELLoG	λ 0.1164 (0.0835)	c 0.9998 (0.1920)	α 5.5095 (6.0111)	δ 0.460 (0.5023)	182.1902	190.1902	191.333	196.9457	0.0661	0.4979	0.1276	0.5323
EHLOW-TL-LLoG	b 3.6755 (1.6715)	β 0.0874 (0.0781)	δ 1.9759 (0.3678)	c 3.6426 (3.0196)	184.4106	192.4106	193.5535	199.1662	0.1112	0.7757	0.1613	0.2484
OEHL-BXII	α 0.3830 (9.5830×10^{-02})	λ 49.8360 (5.6985×10^{-04})	a 8.0611 (3.0917)	b 0.0018 (9.6684×10^{-04})	197.1949	205.1949	206.3377	211.9504	0.1348	0.8164	0.1436	0.3810
APExLLD	α 0.0026 (0.0040)	a 1.7305 (0.3693)	b 24.2383 (0.0019)	c 0.5278 (0.1287)	190.4375	198.4375	199.5803	205.1930	0.1154	0.8056	0.1587	0.2655
APTLW	θ 0.1631 (0.0566)	α 238.4700 (3.6715×10^{-05})	β 1.1866 (0.1655)	λ 0.0630 (0.0327)	199.5483	207.5487	208.6915	214.3042	0.2206	1.5189	0.1857	0.1266

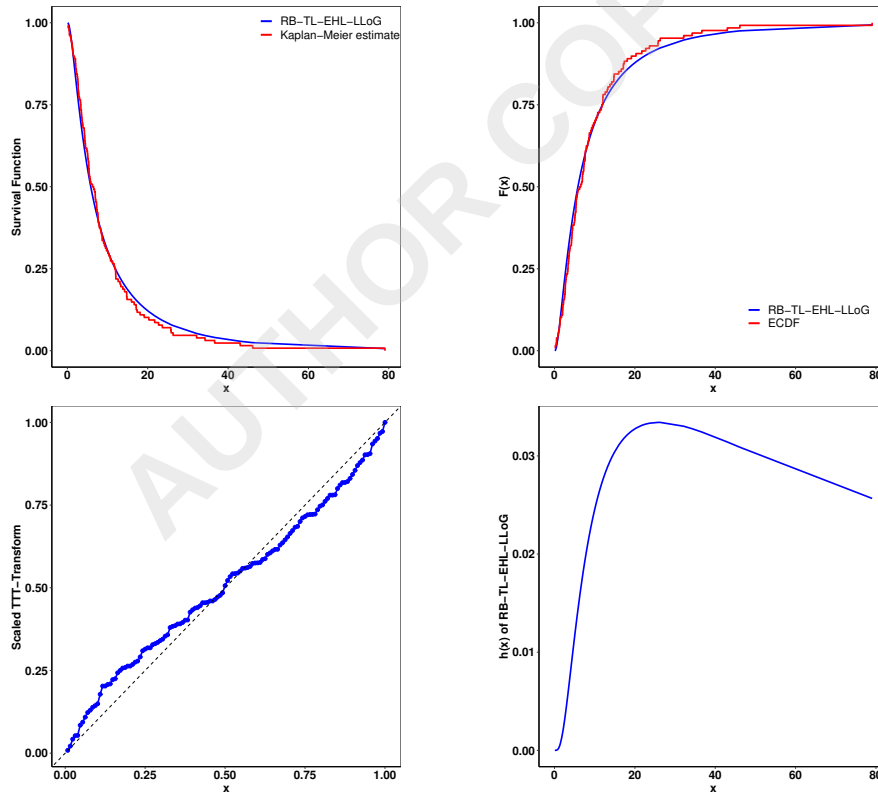


FIGURE 7. Fitted K-M survival curve, theoretical and ECDF, the TTT plot, and plot of hrf for the remission times data

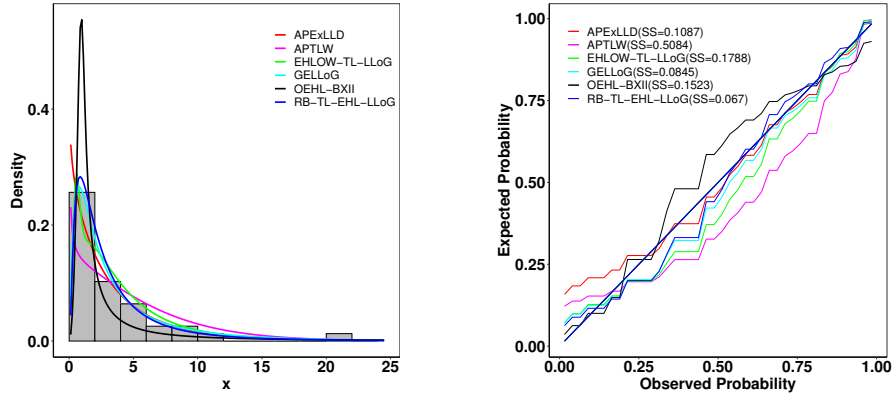


FIGURE 8. Histogram superposed by fitted density (left) and observed vs. expected probability plots (right) for the active repair times data

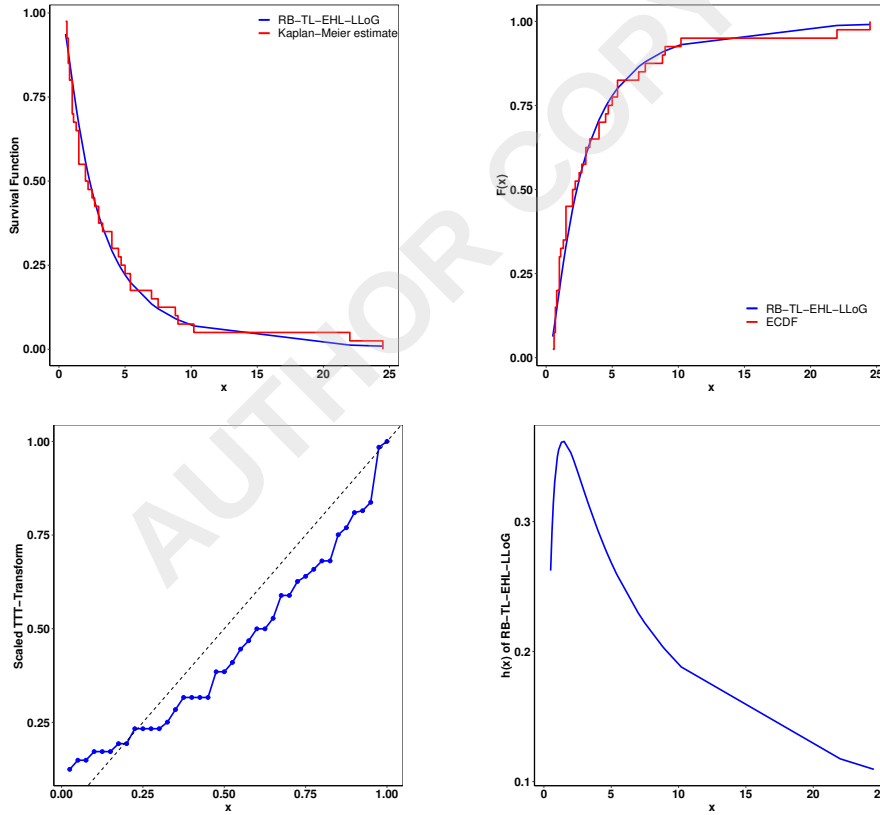


FIGURE 9. Fitted K-M survival curve, theoretical and ECDF, the TTT plot, and the fitted hrf for the active repair times data

9. Conclusion

We have developed a new family of distributions called Ristić-Balakrishnan-Topp-Leone-Exponentiated half logistic-G distributions. By expanding the density function, we were able to obtain important statistical characteristics and estimate the parameters of this family using the maximum likelihood method.

We assessed the accuracy and convergence of the MLE estimates through a Monte Carlo simulation. The flexibility of the Ristić-Balakrishnan-Topp-Leone-Exponentiated half logistic-G family of distributions was demonstrated by fitting it to several real-world data sets.

Supplementary information

Please access the Supplementary Information (pdf 305 KB) on GitHub via the following link:
<https://github.com/shusenpu/SI/blob/main/Supplementary.pdf>.

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*Department of Mathematics and Statistical Sciences
Botswana International University of Science and Technology
Palapye
BOTSWANA
E-mail: oluyedeo@biust.ac.bw
thatayaone.moakofi@studentmail.biust.ac.bw

**Department of Mathematics and Statistics
University of West Florida
Pensacola, FL
USA
E-mail: ls181@students.uwf.edu
acohen@uwf.edu
spu@uwf.edu

***Department of Mathematics and Statistics
University of West Florida
Pensacola, FL
International Baccalaureate Program
Pensacola High School
Pensacola, FL
USA
E-mail: seanfang08@gmail.com

****Department of Applied Physics and Applied Mathematics
Columbia University
New York, NY
USA
E-mail: yz4419@columbia.edu